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Freshwater Forcing of Atlantic Meridional Overturning Circulation Revisited

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Freshwater (FW) forcing is widely identified as the dominant mechanism causing reductions of the Atlantic Meridional Overturning Circulation (AMOC), a climate tipping point that led to past abrupt millennial-scale climate changes. However, the AMOC response to FW forcing has not been rigorously assessed due to the lack of long-term AMOC observations and uncertainties of sea-level rise and ice-sheet melt needed to infer past FW forcing. Here we show a muted AMOC response to FW forcing – ~50-m sea-level rise from the final deglaciation of Northern Hemisphere ice sheets – in the early-to-middle Holocene ~11,700-6,000 years ago. Including this muted AMOC response in a transient simulation of the Holocene with an ocean-atmosphere climate model improves agreement between simulated and proxy temperatures of the past 21,000 years. This demonstrates that the AMOC may not be as sensitive to FW fluxes and Arctic freshening as is currently projected for the end of the 21st century.

Pre-industrial climate evolution of the past 21,000 years indicates that global climate change was paced by Earth's orbital variations and driven mainly by abrupt changes in the Atlantic Meridional Overturning Circulation (AMOC)¹⁻³ and more-gradual changes in atmospheric CO₂^{4,5}. A future disruption of the AMOC⁶ could lead to drying of the Amazon rainforest, disruption of the Asian monsoon⁷, rapid sea-level rise on the northeast coast of the United States⁸, and widespread cessation of crop production in Europe⁹. Indirect assessments of AMOC trends from historical records are inconclusive, with an estimated 15% weakening since the mid-twentieth century based on sea surface temperature (SST) observations¹⁰ and no change in the AMOC state since 1990s based on hydrographic data¹¹. Continuous measurements of the AMOC started in 2004 with the Rapid Climate Change-Meridional Overturning Circulation and Heatflux Array (RAPID) program at 26.5°N, which show that the AMOC weakened between 2004 and 2012 with a recovery since 2012¹². Future projections of the AMOC with climate model simulations under high emission scenarios suggest a best estimate of 34-45% weakening of the AMOC during the 21st century^{13,14} with surface warming and increased freshwater (FW) fluxes to the Arctic and North Atlantic Oceans from runoff and precipitation as well as melting of Arctic sea ice and the Greenland ice sheet¹⁵⁻¹⁷. However, it remains unclear whether the simulated AMOC reduction from global warming is more responsive to changes in surface heat fluxes or FW flux¹⁵ since surface heat flux induces both surface warming and the melting of the Arctic sea ice with liquid FW exports to the North Atlantic Ocean¹⁶, while FW-forcing-only experiments show that an enhanced hydrological cycle¹⁸ and modest increases in FW fluxes projected from Greenland ice-sheet melting^{13,17} can still weaken the AMOC.

While theoretical understanding of the AMOC response to FW forcing is reasonably well established¹⁹, several issues regarding model design and implementation suggest that further

evaluation of this relationship is warranted. For example, a recent study based on hydrographic data suggests a much more stable AMOC than previously thought due to a higher decoupling between the AMOC and ocean interior property fields¹¹. Conversely, another study argues that climate models overestimate AMOC stability due to incorrect net-FW transport in the Atlantic Ocean²⁰, and that an AMOC collapse (~67% reduction) in response to global warming may occur by 2300 after correcting these biases with flux adjustments in the model²¹, with the caveat that a model that correctly simulates surface density does not necessarily correctly simulate stability²². Another issue is the application of FW forcing to regions of deep water formation²³. This includes concerns of correctly producing FW export from the Arctic to the North Atlantic Ocean through boundary currents in low-resolution ocean models²⁴ as well as the time scale and rate of FW forcing that affect the accumulated FW forcing to regions of deep water formation^{25,26}, but previous eddy-permitting ocean simulations (1/6° resolution) with realistic boundary currents show that AMOC reduces by ~30% with 1-year FW forcing from the Arctic²⁴. Further issues include whether climate models form deep water in the correct region²⁷ and properly export FW from the subpolar gyre to the subtropical gyre through the Canary Current²⁸. Finally, we note that multi-model ensemble in Coupled Model Intercomparison Project Phase 6 (CMIP6) show that model resolution in itself does not impact the projected AMOC decline at the end of 21st century^{14,29} and recent studies with eddy-permitting coupled ocean-atmosphere (1/4° resolution) models²⁶ have identified similar rates and magnitudes of AMOC weakening to a 0.1 Sv FW forcing found in traditional non-eddy-permitting models.

Paleoclimate data synthesis

Much of the debate on the response of the AMOC to FW forcing reflects the lack of long-term observations of the AMOC and FW fluxes that could be used to validate models¹⁰⁻¹². In this

regard, the early-to-middle Holocene from ~11,700 to 6,000 years ago (11.7-6.0 ka) provides an opportunity to assess this issue due to well-constrained reconstructions of the AMOC and FW fluxes. In particular, global mean sea level rose ~60 m during this interval, with ~50 m of that rise derived from the final deglaciation of Northern Hemisphere ice sheets (Fig. 1c)³⁰⁻³². This ice-sheet melting resulted in a sustained FW flux of ~0.1 Sv ($1 \text{ Sv} = 10^6 \text{ m}^3 \text{ s}^{-1}$) to the Arctic and North Atlantic Oceans³³⁻³⁶, which is similar to the distribution and amount (0.07-0.12 Sv) of projected runoff and precipitation minus evaporation (P-E) associated with future global warming^{37,38}. In addition, after the opening of the Bering Strait following the Younger Dryas cold interval³⁹, FW transport from the Pacific Ocean to the Arctic Ocean increased and reached the modern day level of ~0.08 Sv³⁸ around 6 ka associated with the ~60 m sea-level rise during this interval. Climate model simulations show that a sustained flux of ~0.10-0.18 Sv should have caused a significant reduction in, if not a collapse of, the AMOC^{14,18,26}.

However, two independent proxies that provide kinematic reconstructions of the AMOC during the early-to-middle Holocene indicate little response to this FW forcing⁴⁰⁻⁴². Instead, the ²³¹Pa/²³⁰Th proxy from multiple Atlantic cores^{40,43} as well as the $\delta^{18}\text{O}$ record from the Florida Straits⁴¹ suggest the AMOC strengthened before ~9 ka and remained at a strength similar to the late Holocene between 9 ka and 6 ka (Fig. 1b). Furthermore, with the exception of the century-long 8,200-yr cold event⁴⁴, and perhaps two other similarly short-lived cold events^{45,46}, reconstructed Holocene Greenland surface temperatures show no signal of an AMOC-induced surface cooling such as the late-Pleistocene Younger Dryas cold period⁴⁷ (Fig. 1a). Instead, Greenland temperatures warmed during the early Holocene and remained at the level similar to the late Holocene between 9 ka and 6 ka in parallel with the AMOC changes suggested by the kinematic proxies^{40,42} (Fig. 1). Having a sustained FW flux of ~0.10-0.18 Sv discharged into the

North Atlantic and Arctic Oceans from 11.7-6.0 ka in association with with little or no slowdown of the AMOC and associated cooling of North Atlantic climate constitutes a fundamental challenge to the paradigm of FW forcing of the AMOC, which we refer to as the “Holocene Meltwater-AMOC Paradox” (HMAP).

Transient Holocene simulations

We next illustrate how overestimation of AMOC sensitivity to FW forcing might cause temperature biases in future projections by comparing two transient simulations of the Holocene with and without FW forcing using the Community Climate System Model version 3 (CCSM3), a coupled ocean-atmosphere climate model of the US National Center for Atmospheric Research. We compare the surface temperature from the two simulations with three regional proxy temperature stacks from Greenland, Antarctica, and the Eastern Atlantic Ocean and Mediterranean Sea area that are known to be strongly influenced by changes in the AMOC (Fig. 2 and Extended Data Figs. 1-5; see Methods for further details).

The original TraCE-21K simulation (herein TraCE-21K-I) was forced by Earth’s orbital variations, greenhouse gases, ice-sheet variations, and FW forcing^{48,49}. Due to large uncertainties of geologic reconstructions of FW forcing before Bølling warming (~14.7 ka), the FW scheme in TraCE-21K-I was designed to reproduce changes in the AMOC as suggested by proxies between the Last Glacial Maximum (~21 ka) and onset of the Bølling warming, followed by a switch to a geologic reconstruction of FW forcing after the onset of Bølling warming^{33-36,49}. For the Holocene, contributions to the FW forcing include the sustained ~0.1 Sv meltwater flux from Northern Hemisphere ice sheets to the Arctic and North Atlantic Ocean in the early-to-middle Holocene (Extended Data Figs. 6-7, Supplementary Table 2) and continuous inflow of fresher North Pacific water to the Arctic and North Atlantic Oceans after the opening of Bering Strait⁴⁹.

The simulated AMOC exhibits good agreement, by design, with proxy reconstructions of the AMOC through the onset of Bølling warming (21.0-14.5 ka) in TraCE-21K-I, with a strong AMOC reduction during the Oldest Dryas⁴⁸ (Fig. 2a). However, under the FW forcing during the Holocene, the AMOC never recovers to the strength suggested by the proxies, being weakest during the period of the HMAP and remaining weak in response to Bering Strait throughflow⁴⁹ in the late Holocene.

Global and hemispheric climate evolution simulated by TraCE-21K-I was in good agreement with global and regional proxy temperature stacks up to and including the onset of Bølling warming^{4,5,48}. However, as with the AMOC, this agreement subsequently breaks down (Fig. 2). In particular, during the period of the HMAP, there is a clear mismatch between the proxy and modeled regional temperature stacks (Fig. 2 b-d and Extended Data Fig. 5), with a >8°C cold bias in central Greenland, a >3°C cold bias in the Eastern Atlantic Ocean and Mediterranean Sea, and an ~2°C warm bias over Antarctica relative to the proxy records. The sign and amplitude of the simulated temperature biases reflects a bipolar seesaw response^{18,48} to the weaker simulated AMOC during the early Holocene (Fig. 2a), which could also produce large biases in tropical precipitation and the global monsoons⁷. Similar temperature-AMOC biases during the early Holocene were also found in transient simulations of the early Holocene from the Loch-Vecode-Ecbilt-Clio-Agism Model (LOVECLIM) and the Fast Met Office/UK Universities Simulator (FAMOUS) model^{50,51}.

We reran TraCE-21K following the onset of Bølling warming (herein TraCE-21K-II) with the same climatic forcing as in TraCE-21K-I but with no FW fluxes during the Bølling-Allerød interstadial (~14.7 ka – 12.9 ka) and throughout the Holocene (Methods). In contrast to the TraCE-21K-I simulation with Holocene FW forcing, the modeled AMOC in TraCE-21K-II is

in better agreement with proxy Holocene AMOC kinematic reconstructions (Fig. 2a). This includes a two-phase recovery suggested by the highest resolution reconstruction from the Florida Straits (blue in Fig. 1b)⁴¹ involving an initial abrupt increase of the modeled AMOC after the end of FW forcing that was prescribed to cause the AMOC reduction during the Younger Dryas (Methods) followed by a further increase to full Holocene rates at ~9 ka (Fig. 2a). Ref.⁵² attributed this two-phase AMOC recovery to an initial increase in deep water formation largely in the North Atlantic subpolar gyre and Irminger Sea regions followed by an abrupt increase of the AMOC when a density threshold is crossed in the Nordic Seas. In addition, the two-phase recovery of the AMOC is responsible for the temperature over Greenland and Eastern Atlantic Ocean/Mediterranean Sea reaching Holocene levels at ~9 ka (Fig 2b, c). Nevertheless, the model does not reproduce the transient temperature evolution during the two-phase recovery in the regional proxy temperature stacks due to the lack of understanding of the physical processes from changes in insolation, ice sheets and atmospheric greenhouse-gas concentrations that are responsible for transient AMOC changes on centennial time scales, although the changes in the latter two forcings during the Holocene likely had a small or negligible effect on the AMOC^{13,51,53}.

The more-realistic simulation of the AMOC after the two-phase recovery substantially improves the agreement between simulated temperatures in TraCE-21K-II and proxy temperatures (Fig. 2 b-d, Extended Data Fig. 5), largely removing the bias of the bipolar seesaw response due to the Holocene AMOC reduction in TraCE-21K-I (Fig. 3). Specifically, the cold bias of >8°C over Greenland and >3°C over the Eastern Atlantic Ocean/Mediterranean Sea in the TraCE-21K-I simulation during the period of the HMAP is reduced by ~80% and ~60%, respectively (Fig. 2b, 2c and Extended Data Fig.5). There is also an ~80% reduction of the warm

bias over Antarctica due to the enhanced northward heat transport from the AMOC in TraCE-21K-II (Fig. 2d and Extended Data Fig. 5).

Implications for past climate changes

The successful transient simulation of the Holocene without FW forcing in TraCE-21K-II supports our inferences from the proxy data synthesis of the muted AMOC response to FW forcing in the Holocene. In addition, prior to the onset of Bølling warming, TraCE-21K-I simulated reasonable surface climate changes using a FW scheme designed to reproduce proxy-based AMOC changes⁴⁸, suggesting that prescribing the reconstructed AMOC instead of reconstructed FW forcing could improve the surface climate simulation of the Holocene and likely other past climate changes. Recent assessments of several proxy-based AMOC reconstructions during the last deglaciation concluded that they show coherent and robust changes during Heinrich event 1 and the Younger Dryas^{41,43}. Nevertheless, these proxy signals can be modulated by other processes^{54,55}, and further work will be needed to reduce remaining uncertainties in the reconstructed AMOC changes if they are to be used as the target for prescribing the AMOC in model simulations.

In addition to the possible decoupling between the AMOC and ocean interior properties¹¹, a key issue likely contributing to the HMAP concerns how the meltwater from Northern Hemisphere ice sheets is distributed to sites of deep-water formation²³. Much of the 0.1 Sv meltwater flux from retreating Northern Hemisphere ice sheets, however, entered the oceans along 1000's of kilometers of coastline bordering those oceans. For example, an estimated 0.02 Sv from the northern Laurentide ice sheet entered the Arctic Ocean along ~2,000 km of coastline during the early Holocene³⁴ which, if evenly distributed, corresponds to 0.0001 Sv per 10 km. Such small, local fluxes would likely be trapped along the coastline and quickly mixed by tides,

wind forcing, and local circulation, and thus unlikely be uniformly spread over sites of North Atlantic deep-water formation.

One question raised by the HMAP is whether the sensitivity of the AMOC to FW forcing differed during the last deglaciation. Unfortunately, however, this question cannot be currently addressed because of the large uncertainties in the FW forcing during the last deglaciation which led to the strategy used by TraCE-21K-I of prescribing FW forcing to cause changes in the AMOC consistent with proxy reconstructions between the Last Glacial Maximum (~21 ka) and onset of the Bølling warming^{48,49}. Whether that FW forcing is realistic and how much of the deglacial AMOC variability may have been associated with other forcings (e.g., ice-sheet orography, insolation) thus remains unclear⁵¹.

Another question raised by the HMAP is whether the cold bias associated with a reduced AMOC in TraCE-21K-I and other transient Holocene simulations⁵⁰ reduced the modeled expression of a Holocene climate optimum around 9,000 to 5,000 years ago⁵⁶ that has been documented extensively in proxy records^{57,58}. We find that the new transient Holocene simulation in TraCE-21K-II exhibits a brief climate optimum at ~9 ka (Extended Data Figs. 8-9) after removing the cold bias in the North Atlantic region in TraCE-21K-I during the HMAP, suggesting the missing Holocene climate optimum in TraCE-21K-I may in part be due to the cold bias from overestimation of AMOC sensitivity to FW forcing during the HMAP. For the late Holocene temperature conundrum⁵⁹, the lack of a cooling trend in TraCE-21K-I has been attributed to the underestimation of Arctic sea ice sensitivity to orbital forcing⁶⁰ and the lack of anthropogenic forcing from the Holocene deforestation in transient Holocene simulations⁶¹.

Implications for future projections

Although the HMAP raises questions about the overestimation of AMOC sensitivity to FW forcing in current climate models, we emphasize that it does not challenge the role of the AMOC in causing abrupt climate changes in the past and potentially in the future. For instance, although the source and magnitude of FW forcing required to slow the AMOC during the Younger Dryas cold interval is still debated^{62,63}, the abrupt decrease of the AMOC during the Younger Dryas is widely accepted as the primary cause of the associated cooling (Fig. 2)^{3,64}. We draw an analogy to atmospheric CO₂ whereby its role in causing past climate change is clear while at the same time there are large uncertainties in our understanding of the physical and biogeochemical processes and feedbacks that caused lower CO₂ during the Last Glacial Maximum and its subsequent increase⁶⁵. Models thus prescribe CO₂ as a forcing of past climate changes using concentrations from ice-core records, whereas the simulated future emission-driven CO₂ changes using carbon-cycle models are regarded as uncertain⁶⁶. For the same reason, we suggest that until the HMAP is resolved, any simulated future AMOC changes from FW forcing and associated temperature, precipitation and regional sea level changes should be viewed with caution (Fig. 3). In particular, having a stable AMOC in the face of sustained and large ~0.10 FW fluxes from ice-sheet melting and ~0.08 Sv from Bering Strait opening during the HMAP suggests that current projections of AMOC decline in the 21st century^{14,66} from projected increase of runoff and P-E^{37,38} as well as the FW exports to the North Atlantic Ocean as the result of the melting of Arctic sea ice from surface warming¹⁶ may be overestimated, which precludes their use for assessing the likelihood of abrupt AMOC changes in the 21st century. As the projected increase of FW input into the Arctic at the end of the 21st century reaches a similar level of ~0.1 Sv FW forcing (~0.05 Sv from runoff, ~0.015 Sv from P-E³⁸, and 0.02-0.04 Sv from melting of the Greenland ice sheet^{17,67-69}) as that associated with early Holocene ice-sheet

245 melting, we conclude that there is an urgent need to assess whether AMOC sensitivity to FW
246 forcing is overestimated in current climate models and investigate alternative mechanisms for
247 past AMOC disruptions in both glacial and interglacial periods and incorporate these
248 mechanisms in climate models for future projections.

249

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Competing interests The authors declare no competing interests.

Additional information

Supplementary information is available for this paper.

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Figure Captions

Fig. 1. Holocene Meltwater-AMOC Paradox. **a**, Composite of temperature reconstruction over the central Greenland based on $\delta^{15}\text{N}$ for 22–10 ka at GISP2, NGRIP and NEEM site⁴⁷ and $\delta^{18}\text{O}$ for 10–0 ka at GISP2 site⁷⁰. **b-c**, AMOC (**b**) and sea-level rise (**c**) during the last deglaciation and Holocene³². The AMOC reconstructions are based on $^{231}\text{Pa}/^{230}\text{Th}$ ratio^{40,42} (green) and cross strait $\delta^{18}\text{O}$ at the Florida Straits⁴¹ (blue). The gray shading highlights the period of the HMAP with muted AMOC response to FW forcing associated with the ~50-m sea-level rise from the final deglaciation of Northern Hemisphere ice sheets. ka, thousand years before 1950.

Fig. 2. Comparison of data and models for regional temperature stacks of past 21,000 years. **a**, AMOC reconstruction from $^{231}\text{Pa}/^{230}\text{Th}$ ratio in Bermuda rise^{40,42} and modeled maximum AMOC transport (below 500 m in the Atlantic Ocean). Sv, Sverdrup ($10^6 \text{ m}^3 \text{ s}^{-1}$). **b-d**, surface air temperature stacks over Greenland (**b**), Antarctica (**d**) and SST stack in the Eastern Atlantic Ocean/Mediterranean Sea (**c**). Proxy data in black; before and include the onset of the Bølling, TraCE-21K-I and TraCE-21K-II are identical (red); after the onset of the Bølling, simulation based on the protocol of prescribing the reconstructed AMOC in red (TraCE-21K-II) and prescribing the reconstructed FW forcing in cyan (TraCE-21K-I). All modeled changes are referenced to the proxy data during the Oldest Dryas (19–15 ka) to aid the comparison. Note that scaling of modeled AMOC versus the $^{231}\text{Pa}/^{230}\text{Th}$ ratio is only intended to capture the relative range of variability. LGM, last glacial maximum. OD, Oldest Dryas. BA, Bølling-Allerød interstadial. YD, Younger Dryas. HMAP, Holocene Meltwater-AMOC paradox. LH, late Holocene. ka, thousand years before 1950.

Fig. 3. Differences of modeled surface temperature and precipitation during the early Holocene (9 ka – 6 ka) between simulations with and without FW forcing in the Holocene. The pattern of the temperature ($^{\circ}\text{C}$) and precipitation (mm/year) differences resembles the classic bipolar seesaw pattern with cooling over Greenland and the Eastern Atlantic Ocean/Mediterranean Sea, warming over Antarctica and southward movement of the Intertropical Convergence Zone (ITCZ) due to the reduction of the northward heat transports with the weaker AMOC in TraCE-21K-I^{18,48}.

References:

- 1 Broecker, W. S., Peteet, D. M. & Rind, D. Does the Ocean-Atmosphere System Have More Than One Stable Mode of Operation. *Nature* **315**, 21-26 (1985).
- 2 Rahmstorf, S. Bifurcations of the Atlantic Thermohaline Circulation in Response to Changes in the Hydrological Cycle. *Nature* **378**, 145-149 (1995).
- 3 Clark, P. U. *et al.* Freshwater forcing of abrupt climate change during the last glaciation. *Science* **293**, 283-287 (2001).
- 4 Shakun, J. D. *et al.* Global warming preceded by increasing carbon dioxide concentrations during the last deglaciation. *Nature* **484**, 49-54 (2012).
- 5 He, F. *et al.* Northern Hemisphere forcing of Southern Hemisphere climate during the last deglaciation. *Nature* **494**, 81-85 (2013).
- 6 Lenton, T. M. *et al.* Climate tipping points - too risky to bet against. *Nature* **575**, 592-595 (2019).
- 7 Jackson, L. C. *et al.* Global and European climate impacts of a slowdown of the AMOC in a high resolution GCM. *Climate Dynamics* **45**, 3299-3316 (2015).
- 8 Yin, J., Schlesinger, M. E. & Stouffer, R. J. Model projections of rapid sea-level rise on the northeast coast of the United States. *Nature Geoscience* **2**, 262-266 (2009).
- 9 Ritchie, P. D. L. *et al.* Shifts in national land use and food production in Great Britain after a climate tipping point. *Nature Food* **1**, 76-83 (2020).
- 10 Caesar, L., Rahmstorf, S., Robinson, A., Feulner, G. & Saba, V. Observed fingerprint of a weakening Atlantic Ocean overturning circulation. *Nature* **556**, 191-196 (2018).
- 11 Fu, Y., Li, F. L., Karstensen, J. & Wang, C. Z. A stable Atlantic Meridional Overturning Circulation in a changing North Atlantic Ocean since the 1990s. *Science Advances* **6**, eabc7836 (2020).
- 12 Moat, B. *et al.* Pending recovery in the strength of the meridional overturning circulation at 26 degrees N. *Ocean Science* **16**, 863-874 (2020).
- 13 Bakker, P. *et al.* Fate of the Atlantic Meridional Overturning Circulation: Strong decline under continued warming and Greenland melting. *Geophysical Research Letters* **43**, 12252-12260 (2016).

331 14 Weijer, W., Cheng, W., Garuba, O. A., Hu, A. & Nadiga, B. T. CMIP6 Models Predict
332 Significant 21st Century Decline of the Atlantic Meridional Overturning Circulation.
333 *Geophysical Research Letters* **47**, e2019GL086075 (2020).

334 15 Gregory, J. M. *et al.* A model intercomparison of changes in the Atlantic thermohaline
335 circulation in response to increasing atmospheric CO₂ concentration. *Geophysical*
336 *Research Letters* **32**, n/a-n/a (2005).

337 16 Jahn, A. & Holland, M. M. Implications of Arctic sea ice changes for North Atlantic deep
338 convection and the meridional overturning circulation in CCSM4-CMIP5 simulations.
339 *Geophysical Research Letters* **40**, 1206-1211 (2013).

340 17 Golledge, N. R. *et al.* Global environmental consequences of twenty-first-century ice-
341 sheet melt. *Nature* **566**, 65-72 (2019).

342 18 Stouffer, R. J. *et al.* Investigating the causes of the response of the thermohaline
343 circulation to past and future climate changes. *Journal of Climate* **19**, 1365-1387 (2006).

344 19 Weijer, W. *et al.* Stability of the Atlantic Meridional Overturning Circulation: A Review
345 and Synthesis. *Journal of Geophysical Research-Oceans* **124**, 5336-5375 (2019).

346 20 Rahmstorf, S. On the freshwater forcing and transport of the Atlantic thermohaline
347 circulation. *Climate Dynamics* **12**, 799-811 (1996).

348 21 Liu, W., Xie, S. P., Liu, Z. & Zhu, J. Overlooked possibility of a collapsed Atlantic
349 Meridional Overturning Circulation in warming climate. *Sci Adv* **3**, e1601666 (2017).

350 22 Gnanadesikan, A., Kelson, R. & Sten, M. Flux Correction and Overturning Stability:
351 Insights from a Dynamical Box Model. *Journal of Climate* **31**, 9335-9350 (2018).

352 23 Wunsch, C. Towards understanding the Paleocean. *Quaternary Science Reviews* **29**,
353 1960-1967 (2010).

354 24 Condrón, A. & Winsor, P. Meltwater routing and the Younger Dryas. *Proc Natl Acad Sci*
355 *U S A* **109**, 19928-19933 (2012).

356 25 Kim, H.-J., An, S.-I., Kim, S.-K. & Park, J.-H. Feedback Processes Modulating the
357 Sensitivity of Atlantic Thermohaline Circulation to Freshwater Forcing Timescales.
358 *Journal of Climate* **34**, 5081-5092 (2021).

359 26 Jackson, L. C. & Wood, R. A. Hysteresis and Resilience of the AMOC in an Eddy-
360 Permitting GCM. *Geophysical Research Letters* **45**, 8547-8556 (2018).

- 361 27 Lozier, M. S. *et al.* A sea change in our view of overturning in the subpolar North
362 Atlantic. *Science* **363**, 516-521 (2019).
- 363 28 Swingedouw, D. *et al.* Decadal fingerprints of freshwater discharge around Greenland in
364 a multi-model ensemble. *Climate Dynamics* **41**, 695-720 (2012).
- 365 29 Jackson, L. C. *et al.* Impact of ocean resolution and mean state on the rate of AMOC
366 weakening. *Climate Dynamics* **55**, 1711-1732 (2020).
- 367 30 Ullman, D. J. *et al.* Final Laurentide ice-sheet deglaciation and Holocene climate-sea
368 level change. *Quaternary Science Reviews* **152**, 49-59 (2016).
- 369 31 Cuzzone, J. K. *et al.* Final deglaciation of the Scandinavian Ice Sheet and implications for
370 the Holocene global sea-level budget. *Earth and Planetary Science Letters* **448**, 34-41
371 (2016).
- 372 32 Lambeck, K., Rouby, H., Purcell, A., Sun, Y. & Sambridge, M. Sea level and global ice
373 volumes from the Last Glacial Maximum to the Holocene. *Proc Natl Acad Sci U S A* **111**,
374 15296-15303 (2014).
- 375 33 Licciardi, J. M., Clark, P. U., Jenson, J. W. & Macayeal, D. R. Deglaciation of a soft-
376 bedded Laurentide Ice Sheet. *Quaternary Science Reviews* **17**, 427-448 (1998).
- 377 34 Licciardi, J. M., Teller, J. T. & Clark, P. U. in *Mechanisms of Global Climate Change at*
378 *Millennial Time Scales* Vol. 112 *Geophysical Monograph Series* 177-201 (AGU, 1999).
- 379 35 Carlson, A. E. *et al.* Rapid early Holocene deglaciation of the Laurentide ice sheet.
380 *Nature Geoscience* **1**, 620-624 (2008).
- 381 36 Siegert, M. J. & Dowdeswell, J. A. Numerical reconstructions of the Eurasian Ice Sheet
382 and climate during the Late Weichselian. *Quaternary Science Reviews* **23**, 1273-1283
383 (2004).
- 384 37 Dixon, K. W., Delworth, T. L., Spelman, M. J. & Stouffer, R. J. The influence of
385 transient surface fluxes on North Atlantic overturning in a coupled GCM climate change
386 experiment. *Geophysical Research Letters* **26**, 2749-2752 (1999).
- 387 38 Haine, T. W. N. *et al.* Arctic freshwater export: Status, mechanisms, and prospects.
388 *Global and Planetary Change* **125**, 13-35 (2015).
- 389 39 Clark, J., Mitrovica, J. X. & Alder, J. Coastal paleogeography of the California–Oregon–
390 Washington and Bering Sea continental shelves during the latest Pleistocene and

391 Holocene: implications for the archaeological record. *Journal of Archaeological Science*
392 **52**, 12-23 (2014).

393 40 Lippold, J. *et al.* Constraining the Variability of the Atlantic Meridional Overturning
394 Circulation During the Holocene. *Geophysical Research Letters* **46**, 11338-11346 (2019).

395 41 Lynch-Stieglitz, J. The Atlantic Meridional Overturning Circulation and Abrupt Climate
396 Change. *Ann Rev Mar Sci* **9**, 83-104 (2017).

397 42 McManus, J. F., Francois, R., Gherardi, J. M., Keigwin, L. D. & Brown-Leger, S.
398 Collapse and rapid resumption of Atlantic meridional circulation linked to deglacial
399 climate changes. *Nature* **428**, 834-837 (2004).

400 43 Ng, H. C. *et al.* Coherent deglacial changes in western Atlantic Ocean circulation. *Nat*
401 *Commun* **9**, 2947 (2018).

402 44 Alley, R. B. *et al.* Holocene climatic instability: A prominent, widespread event 8200 yr
403 ago. *Geology* **25**, 483-486 (1997).

404 45 Fleitmann, D. *et al.* Evidence for a widespread climatic anomaly at around 9.2 ka before
405 present. *Paleoceanography* **23**, n/a-n/a (2008).

406 46 Bond, G. *et al.* A pervasive millennial-scale cycle in North Atlantic Holocene and glacial
407 climates. *Science* **278**, 1257-1266 (1997).

408 47 Buizert, C. *et al.* Greenland temperature response to climate forcing during the last
409 deglaciation. *Science* **345**, 1177-1180 (2014).

410 48 Liu, Z. *et al.* Transient simulation of last deglaciation with a new mechanism for Bolling-
411 Allerod warming. *Science* **325**, 310-314 (2009).

412 49 He, F. *Simulating transient climate evolution of the last deglaciation with CCSM3* Ph.D.
413 thesis, The University of Wisconsin - Madison, (2011).

414 50 Zhang, Y., Renssen, H., Seppä, H. & Valdes, P. J. Holocene temperature trends in the
415 extratropical Northern Hemisphere based on inter-model comparisons. *Journal of*
416 *Quaternary Science* **33**, 464-476 (2018).

417 51 Renssen, H. *et al.* The spatial and temporal complexity of the Holocene thermal
418 maximum. *Nature Geoscience* **2**, 411-414 (2009).

419 52 Renold, M., Raible, C. C., Yoshimori, M. & Stocker, T. F. Simulated resumption of the
420 North Atlantic meridional overturning circulation – Slow basin-wide advection and
421 abrupt local convection. *Quaternary Science Reviews* **29**, 101-112 (2010).

422 53 Zhu, J., Liu, Z. Y., Zhang, X., Eisenman, I. & Liu, W. Linear weakening of the AMOC in
423 response to receding glacial ice sheets in CCSM3. *Geophysical Research Letters* **41**,
424 6252-6258 (2014).

425 54 Du, J., Haley, B. A. & Mix, A. C. Evolution of the Global Overturning Circulation since
426 the Last Glacial Maximum based on marine authigenic neodymium isotopes. *Quaternary*
427 *Science Reviews* **241**, 106396 (2020).

428 55 Missiaen, L. *et al.* Modelling the impact of biogenic particle flux intensity and
429 composition on sedimentary Pa/Th. *Quaternary Science Reviews* **240**, 106394 (2020).

430 56 Bova, S., Rosenthal, Y., Liu, Z., Godad, S. P. & Yan, M. Seasonal origin of the thermal
431 maxima at the Holocene and the last interglacial. *Nature* **589**, 548-553 (2021).

432 57 Kaufman, D. S. *et al.* Holocene thermal maximum in the western Arctic (0–180°W).
433 *Quaternary Science Reviews* **23**, 529-560 (2004).

434 58 Marcott, S. A., Shakun, J. D., Clark, P. U. & Mix, A. C. A reconstruction of regional and
435 global temperature for the past 11,300 years. *Science* **339**, 1198-1201 (2013).

436 59 Liu, Z. *et al.* The Holocene temperature conundrum. *Proc Natl Acad Sci U S A* **111**,
437 E3501-3505 (2014).

438 60 Park, H.-S., Kim, S.-J., Stewart, A. L., Son, S.-W. & Seo, K.-H. Mid-Holocene Northern
439 Hemisphere warming driven by Arctic amplification. *Science Advances* **5**, eaax8203
440 (2019).

441 61 He, F. *et al.* Simulating global and local surface temperature changes due to Holocene
442 anthropogenic land cover change. *Geophysical Research Letters* **41**, 623-631 (2014).

443 62 Carlson, A. E. & Clark, P. U. Ice sheet sources of sea level rise and freshwater discharge
444 during the last deglaciation. *Reviews of Geophysics* **50**, RG4007 (2012).

445 63 Broecker, W. S. Geology. Was the Younger Dryas triggered by a flood? *Science* **312**,
446 1146-1148 (2006).

447 64 Rahmstorf, S. Ocean circulation and climate during the past 120,000 years. *Nature* **419**,
448 207-214 (2002).

- 449 65 Sigman, D. M., Hain, M. P. & Haug, G. H. The polar ocean and glacial cycles in
450 atmospheric CO(2) concentration. *Nature* **466**, 47-55 (2010).
- 451 66 Collins, M. *et al.* Long-term climate change: projections, commitments and
452 irreversibility. In: *Climate Change 2013: The Physical Science Basis. Contribution of*
453 *Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on*
454 *Climate Change* [Stocker, T.F., D. Qin, G.-K. Plattner, M. Tignor, S.K. Allen, J.
455 Boschung, A. Nauels, Y. Xia, V. Bex and P.M. Midgley (eds.)]. (2013).
- 456 67 Church, J. A. *et al.* Sea Level Change. In: *Climate Change 2013: The Physical Science*
457 *Basis. Contribution of Working Group I to the Fifth Assessment Report of the*
458 *Intergovernmental Panel on Climate Change* [Stocker, T.F., D. Qin, G.-K. Plattner, M.
459 Tignor, S.K. Allen, J. Boschung, A. Nauels, Y. Xia, V. Bex and P.M. Midgley (eds.)].
460 (Cambridge University Press, 2013).
- 461 68 Briner, J. P. *et al.* Rate of mass loss from the Greenland Ice Sheet will exceed Holocene
462 values this century. *Nature* **586**, 70-+ (2020).
- 463 69 Aschwanden, A. *et al.* Contribution of the Greenland Ice Sheet to sea level over the next
464 millennium. *Sci Adv* **5**, eaav9396 (2019).
- 465 70 Alley, R. B. The Younger Dryas cold interval as viewed from central Greenland.
466 *Quaternary Science Reviews* **19**, 213-226 (2000).
467

Methods

Transient climate modelling

We conducted version II of the TraCE-21K simulation (TraCE-21K-II) at the Computational and Information Systems Laboratory^{71,72} of National Center for Atmospheric Research with the Community Climate System Model version 3 (CCSM3) based on the protocol of prescribing the reconstructed AMOC instead of the reconstructed FW forcing, which removes the temperature and precipitation biases in the Holocene segment of the first transient simulation of past 21,000 years (TraCE-21K-I)^{48,49} due to the overestimation of the response of AMOC to freshwater fluxes in the coupled global climate models. TraCE-21K-II was conducted with the same climatic forcing from Earth's orbital variations and greenhouse gases as well as similar ice sheets variations as in TraCE-21K-I⁴⁹, but with no freshwater flux being prescribed when the reconstructed AMOC exhibits typical interglacial strength during the Bølling-Allerød interstadial (~14.7 ka – 12.9 ka) and throughout the Holocene from ~11,500 years ago to present day.

Between the LGM and the onset of the Bølling (~14.7 ka), TraCE-21K-I and TraCE-21K-II are identical. TraCE-21K-II was first branched off from the 14.9 ka (ka, thousand years before 1950) of TraCE-21K-I with meltwater fluxes being totally cut off at the onset of Bølling warming around 14.7 ka. Between Bølling warming and the Younger Dryas, no meltwater flux was applied in TraCE-21K-II, which has been demonstrated to be able to produce a reasonable simulation of both Bølling interstadial over Greenland and the Antarctic Cold Reversal over Antarctica and the Southern Ocean⁷³. During the Younger Dryas, a meltwater flux of 0.17 Sv was applied in the North Atlantic Ocean (50°N-70°N) between 12.9 ka and 11.7 ka in TraCE-21K-II to slow the AMOC down to the minimum of ~4 Sv in CCSM3 (Extended Data Figs. 6-7

and Supplementary Table 3) and produce reasonable simulation of cooling over the Greenland and Eastern Atlantic/Mediterranean region⁴⁷ (Fig. 2). At the end of the Younger Dryas, the meltwater forcing was ceased at 11.7 ka and no further meltwater was applied in TraCE-21K-II throughout the Holocene. As a result, the AMOC in TraCE-21K-II during the HMAP is much stronger than that in TraCE-21K-I due to stronger deep water formation associated with deeper winter mix layer depth in TraCE-21K-II (Extended Data Fig. 10). Note that for the Holocene segment of the TraCE-21K-I experiment, contributions to the FW forcing include the sustained ~0.1 Sv meltwater flux from Northern Hemisphere ice sheets and inflow of fresher North Pacific water to the North Atlantic associated with the opening of Bering Strait. After 6 ka, the ~0.1 Sv meltwater flux from Northern Hemisphere ice sheets ended, but the inflow of fresher North Pacific water to the North Atlantic associated with the opening of Bering Strait continued. In order to conduct TraCE-21K-II with no freshwater flux throughout the Holocene, the Bering Strait was kept closed to prevent the inflow of fresher North Pacific water to the North Atlantic, which thus explains the difference between TraCE-21K-II and TraCE-21K-I after 6 ka. As in TraCE-21K-I, TraCE-21K-II includes dynamic vegetation feedback and a fixed annual cycle of aerosol forcing⁴⁹.

Regional temperature stacks

The ice-core and Eastern Atlantic/Mediterranean regional temperature stacks were derived from the deglacial proxy record compilations^{4,47} that contain most published high-resolution (median resolution = 200 years), well-dated (636 radiocarbon dates) temperature records from the last deglaciation, and as such, represent the current state of knowledge on deglacial temperature variability in those regions. The data were linearly interpolated to 100-year resolution and combined as averages to yield mean temperature time series for the regional

temperature stacks. The detailed information of all proxy data for regional temperature stacks is documented in Supplementary Table 1 with the locations plotted in Extended Data Fig. 1. All modeled regional temperature stacks are derived as averages of simulated annual mean temperature with 10-year averages at proxy site locations. Figure 2 shows the comparison between the model and data for regional temperature stacks of past 21,000 years with changes in modeled regional temperature stacks referenced to the averages of proxy regional temperature stacks during the Oldest Dryas (19-15 ka). Extended Data Fig. 4 shows the comparison of regional temperature stacks between data and models with temperature anomalies of each regional stack from the average value in the Oldest Dryas (19 ka-15 ka).

Data Availability

The model datasets used for this study (Figs. 2-3) are available from the Open Science Framework (<https://doi.org/10.17605/OSF.IO/NUQ2K>). TraCE-21K-I model data are available from the Earth System Grid <https://www.earthsystemgrid.org/project/trace.html> and TraCE-21K-II model data are available from the Transient Climate Simulation Lab <https://trace-21k.nelson.wisc.edu>.

Code Availability

CCSM3 is freely available as open-source code from <http://www.cesm.ucar.edu/models/ccsm3.0/>

Methods Reference:

- 71 Computational Information Systems Laboratory. Yellowstone: IBM iDataPlex System
(University Community Computing). Boulder, CO: National Center for Atmospheric
Research. <http://n2t.net/ark:/85065/d7wd3xhc>. (2016).
- 72 Computational Information Systems Laboratory. Cheyenne: HPE/SGI ICE XA System
(University Community Computing). Boulder, CO: National Center for Atmospheric
Research. doi:10.5065/D6RX99HX. (2019).

537 73 Pedro, J. B. *et al.* The spatial extent and dynamics of the Antarctic Cold Reversal. *Nature*
538 *Geoscience* **9**, 51-55 (2015).

539 74 Rasmussen, S. O. *et al.* A new Greenland ice core chronology for the last glacial
540 termination. *Journal of Geophysical Research* **111** (2006).

541 75 Andersen, K. K. *et al.* The Greenland Ice Core Chronology 2005, 15–42ka. Part 1:
542 constructing the time scale. *Quaternary Science Reviews* **25**, 3246-3257 (2006).

543 76 Waelbroeck, C. *et al.* The timing of the last deglaciation in North Atlantic climate
544 records. *Nature* **412**, 724-727 (2001).

545 77 Cacho, I. *et al.* Variability of the western Mediterranean Sea surface temperature during
546 the last 25,000 years and its connection with the Northern Hemisphere climatic changes.
547 *Paleoceanography* **16**, 40-52 (2001).

548 78 Bard, E., Rostek, F., Turon, J. L. & Gendreau, S. Hydrological impact of heinrich events
549 in the subtropical northeast atlantic. *Science* **289**, 1321-1324 (2000).

550 79 Cacho, I. *et al.* Dansgaard-Oeschger and heinrich event imprints in Alboran Sea
551 paleotemperatures. *Paleoceanography* **14**, 698-705 (1999).

552 80 Castañeda, I. S. *et al.* Millennial-scale sea surface temperature changes in the eastern
553 Mediterranean (Nile River Delta region) over the last 27,000 years. *Paleoceanography* **25**
554 (2010).

555 81 Arz, H. W., Patzold, J., Muller, P. J. & Moammar, M. O. Influence of Northern
556 Hemisphere climate and global sea level rise on the restricted Red Sea marine
557 environment during termination I. *Paleoceanography* **18** (2003).

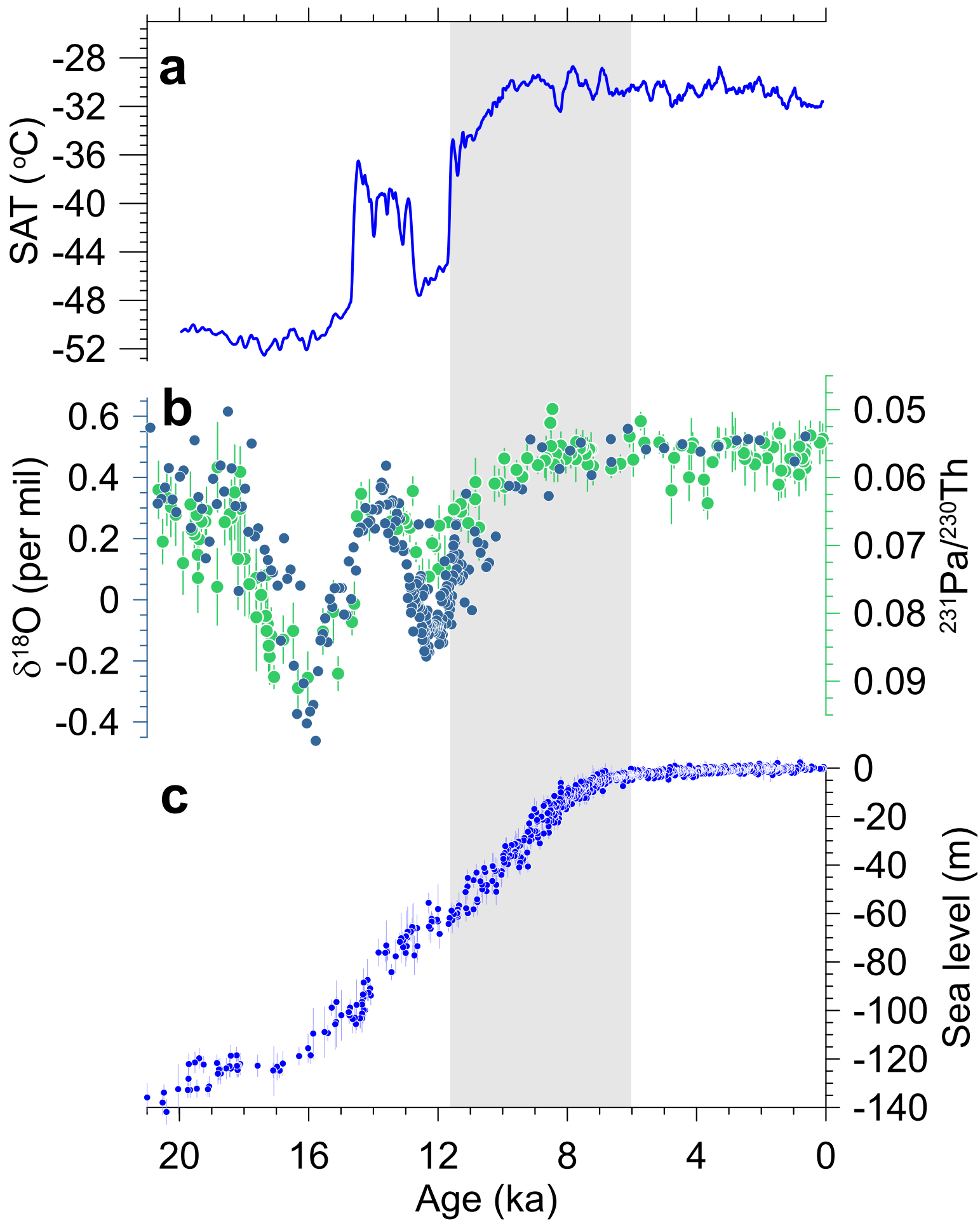
558 82 Stenni, B. *et al.* The deuterium excess records of EPICA Dome C and Dronning Maud
559 Land ice cores (East Antarctica). *Quaternary Science Reviews* **29**, 146-159 (2010).

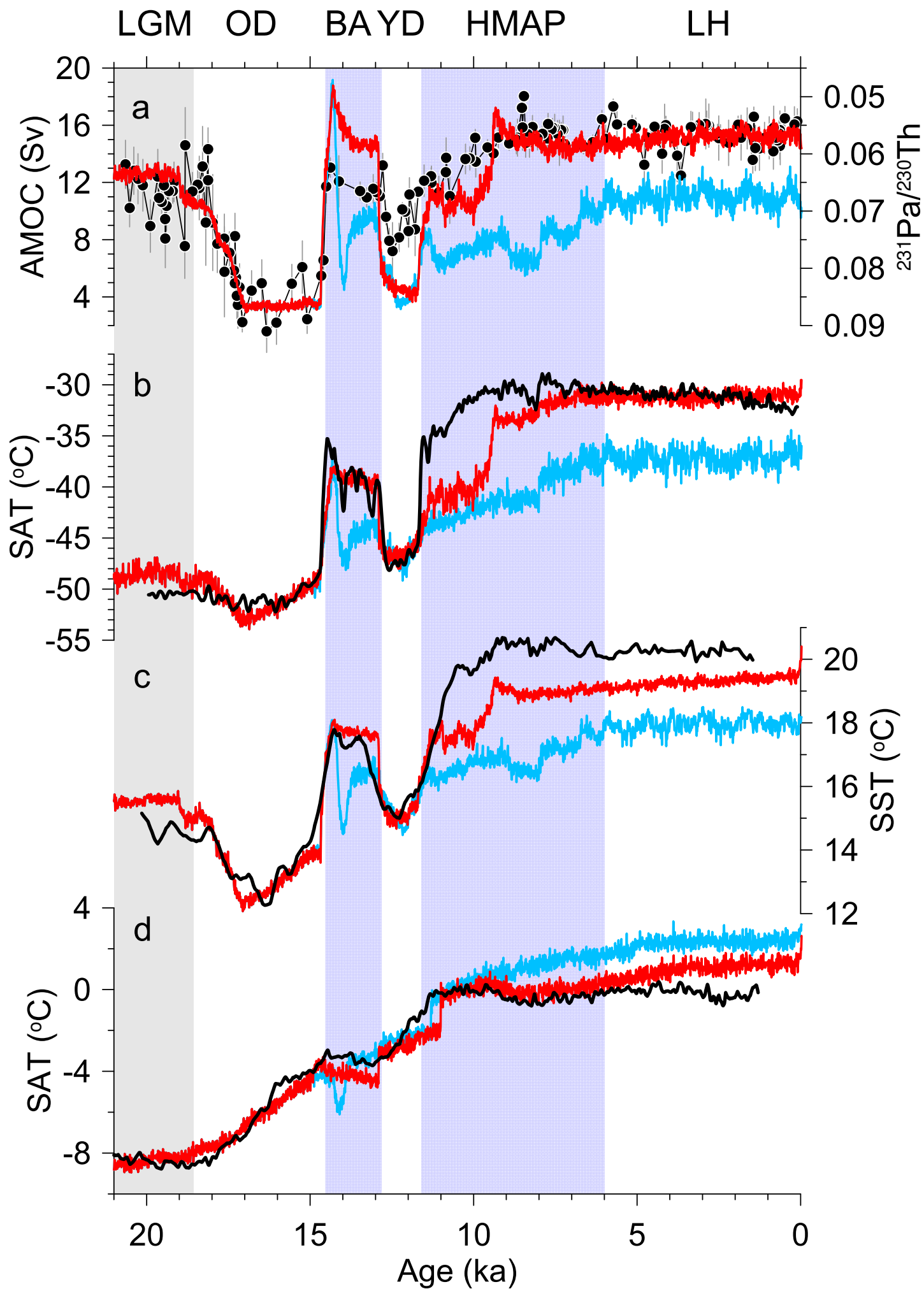
560 83 Lemieux-Dudon, B. *et al.* Consistent dating for Antarctic and Greenland ice cores.
561 *Quaternary Science Reviews* **29**, 8-20 (2010).

562 84 Kawamura, K. *et al.* Northern Hemisphere forcing of climatic cycles in Antarctica over
563 the past 360,000 years. *Nature* **448**, 912-916 (2007).

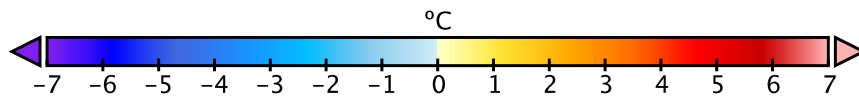
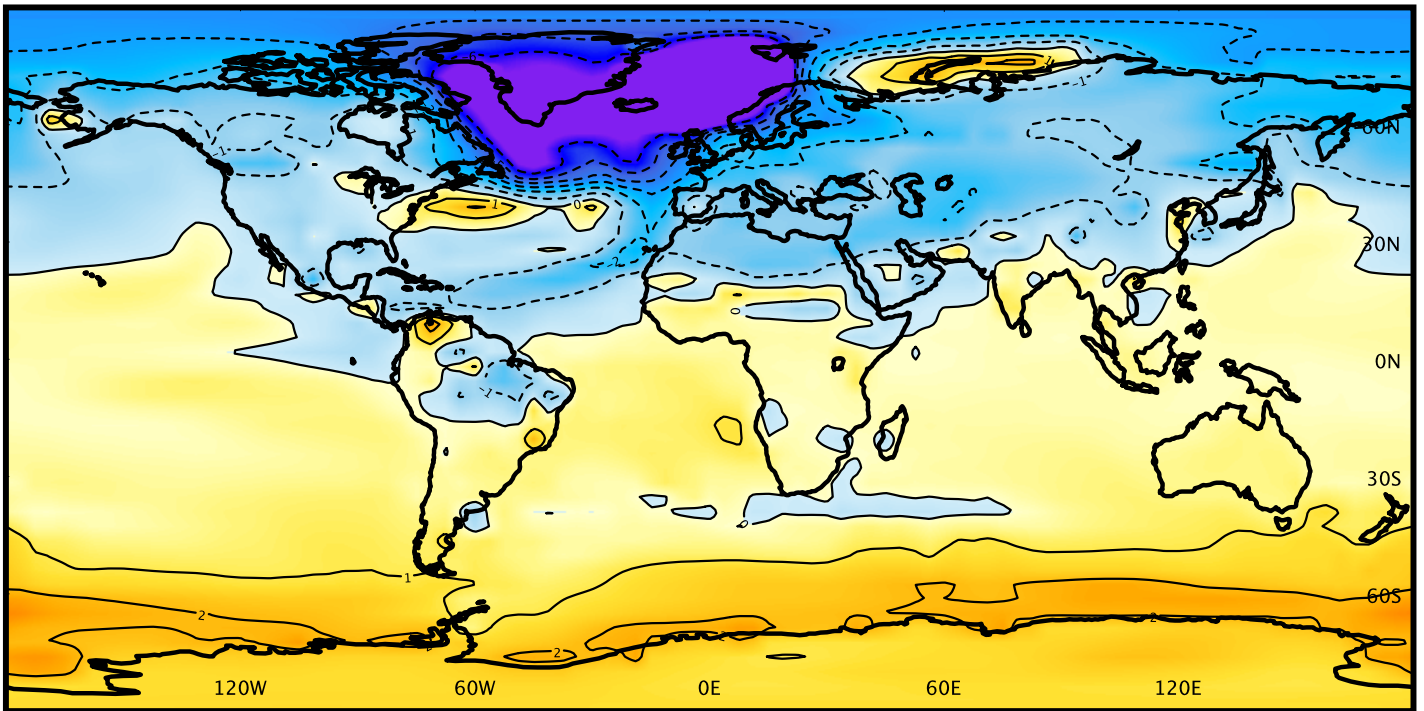
564 85 Petit, J. R. *et al.* Climate and atmospheric history of the past 420,000 years from the
565 Vostok ice core, Antarctica. *Nature* **399**, 429-436 (1999).

566

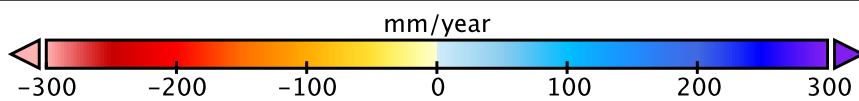
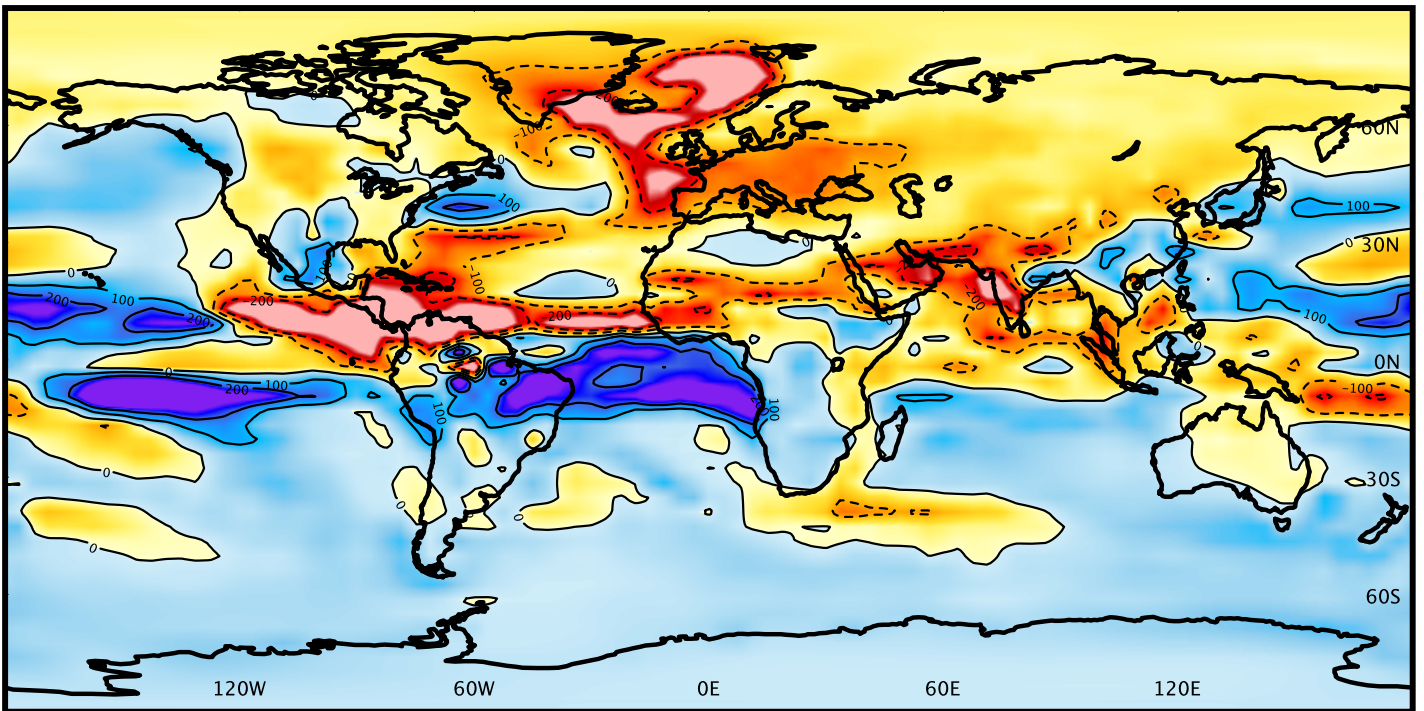


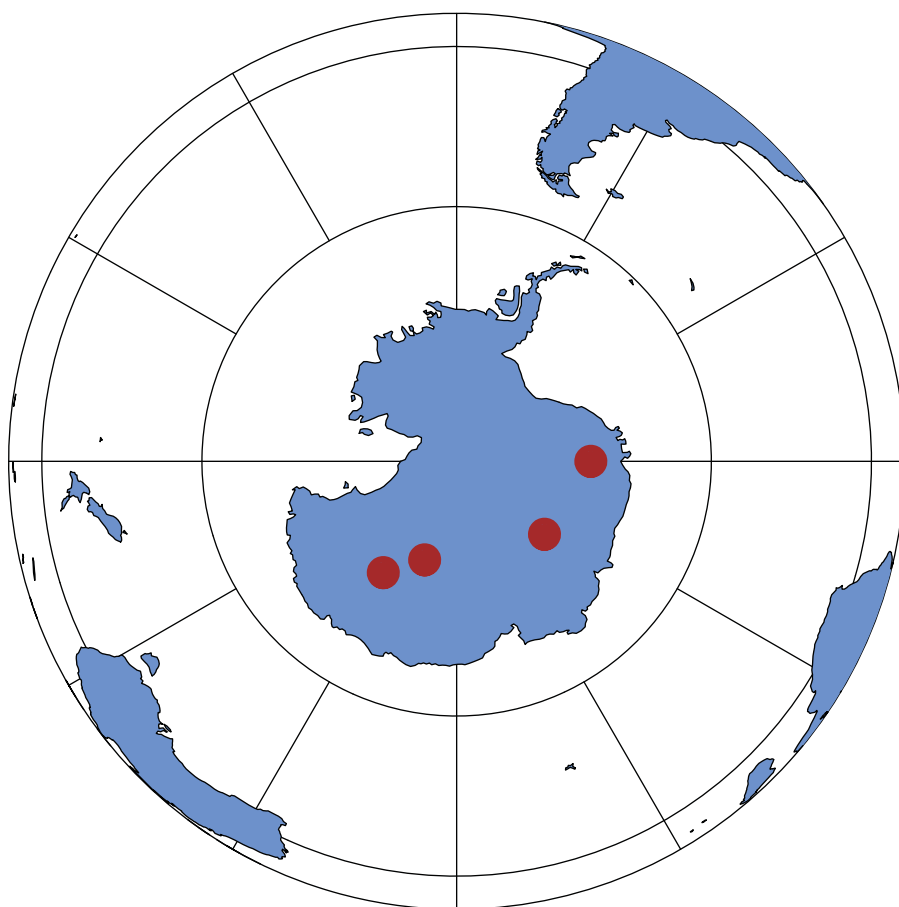


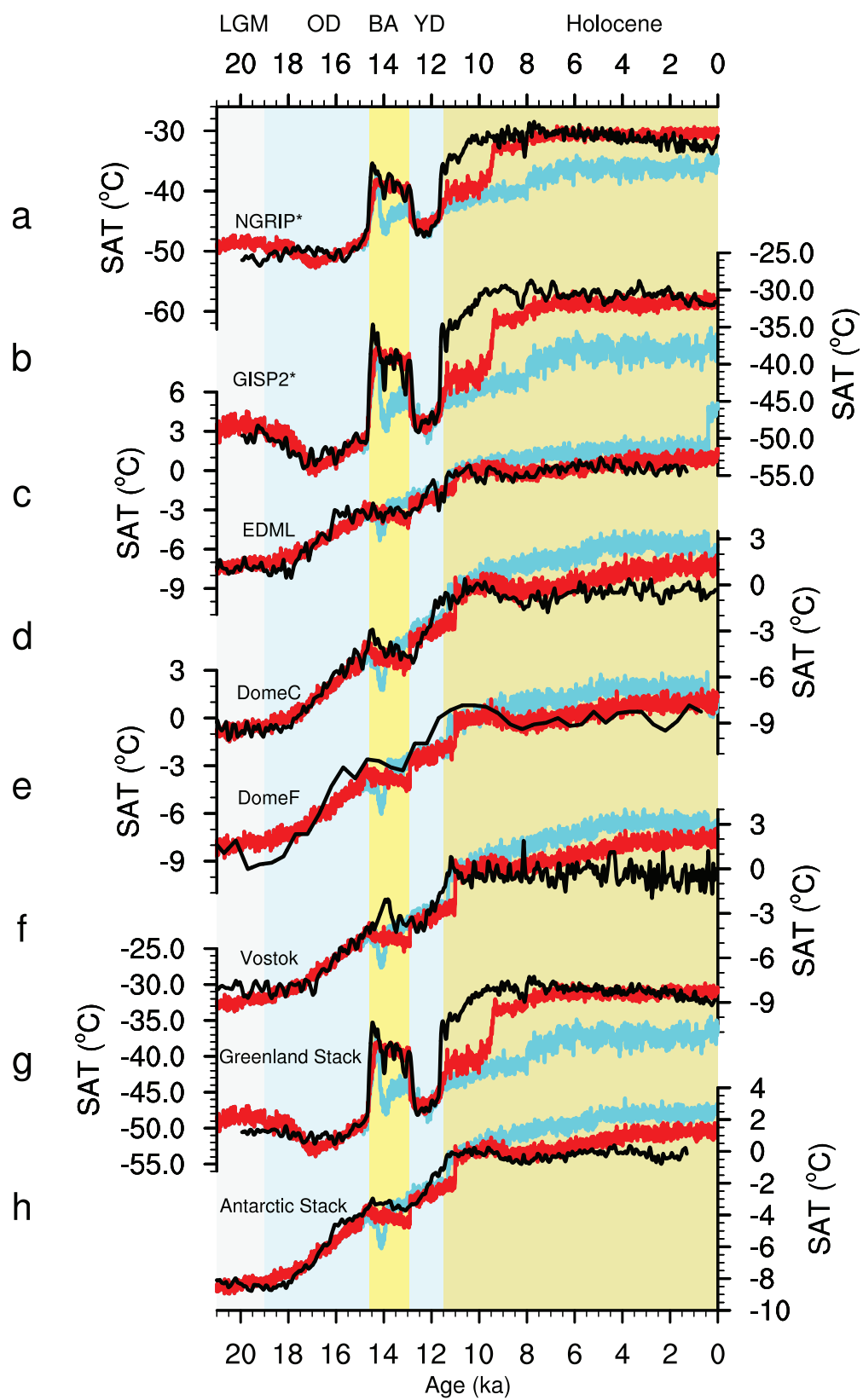
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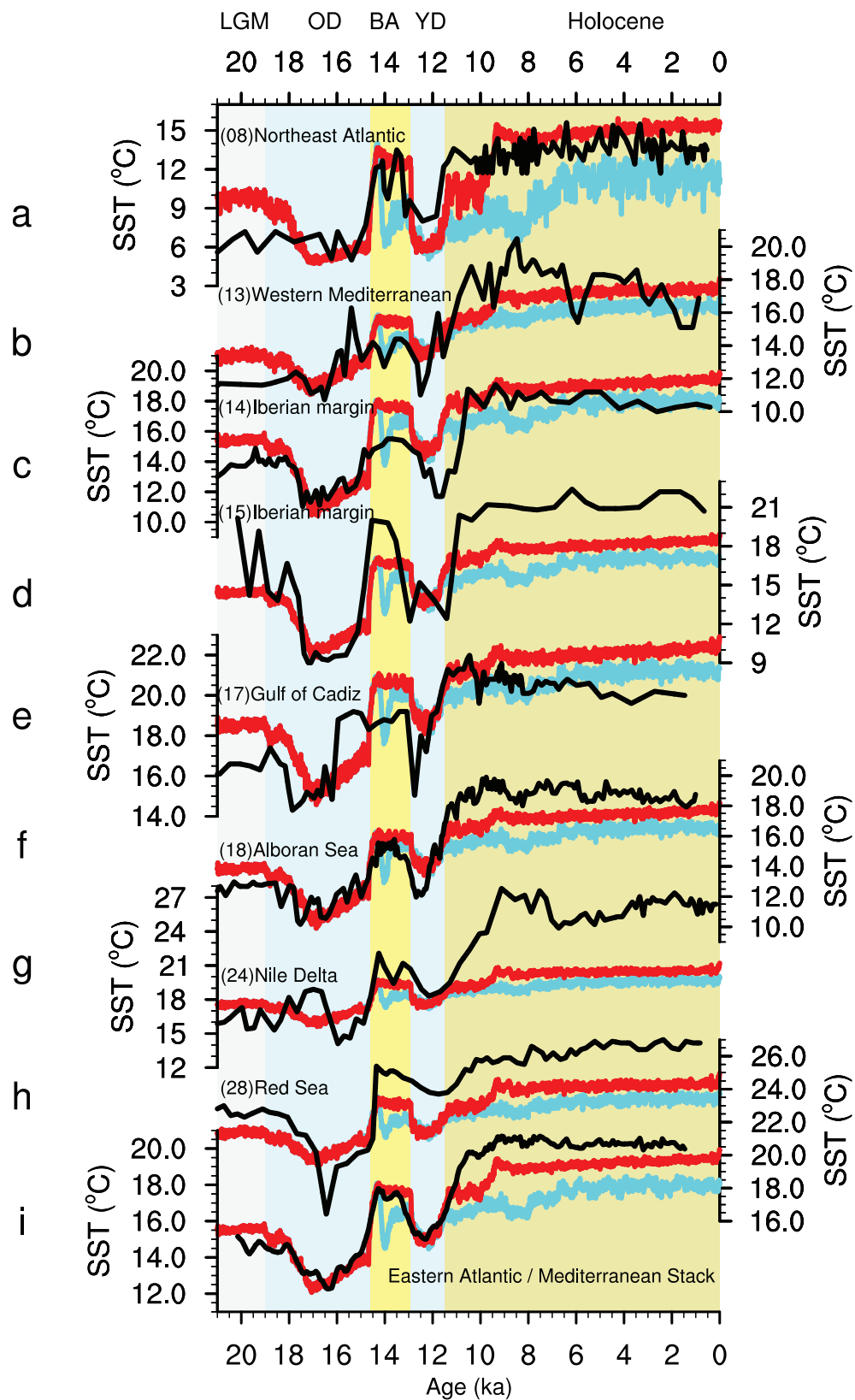


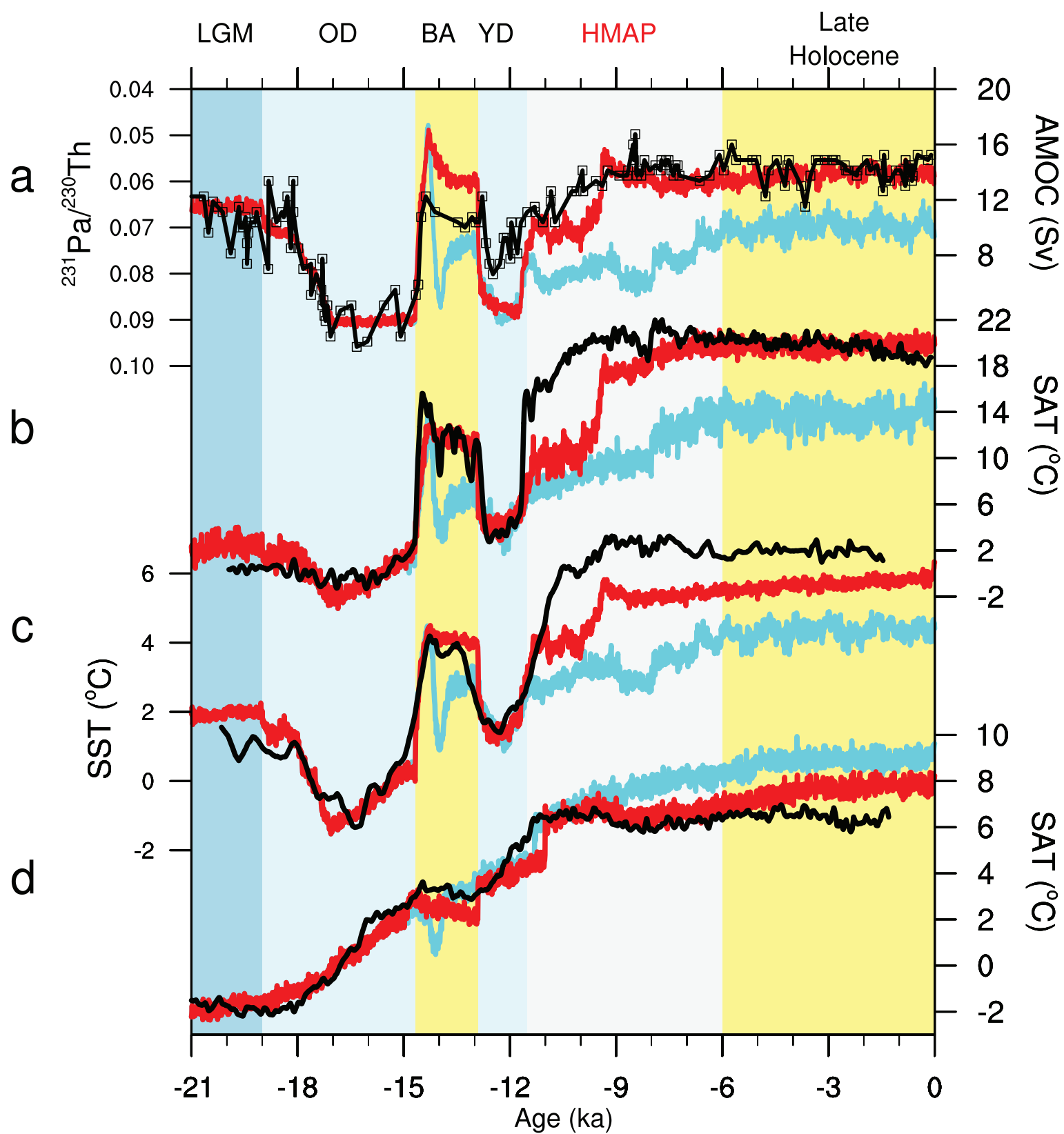
Precipitation rate

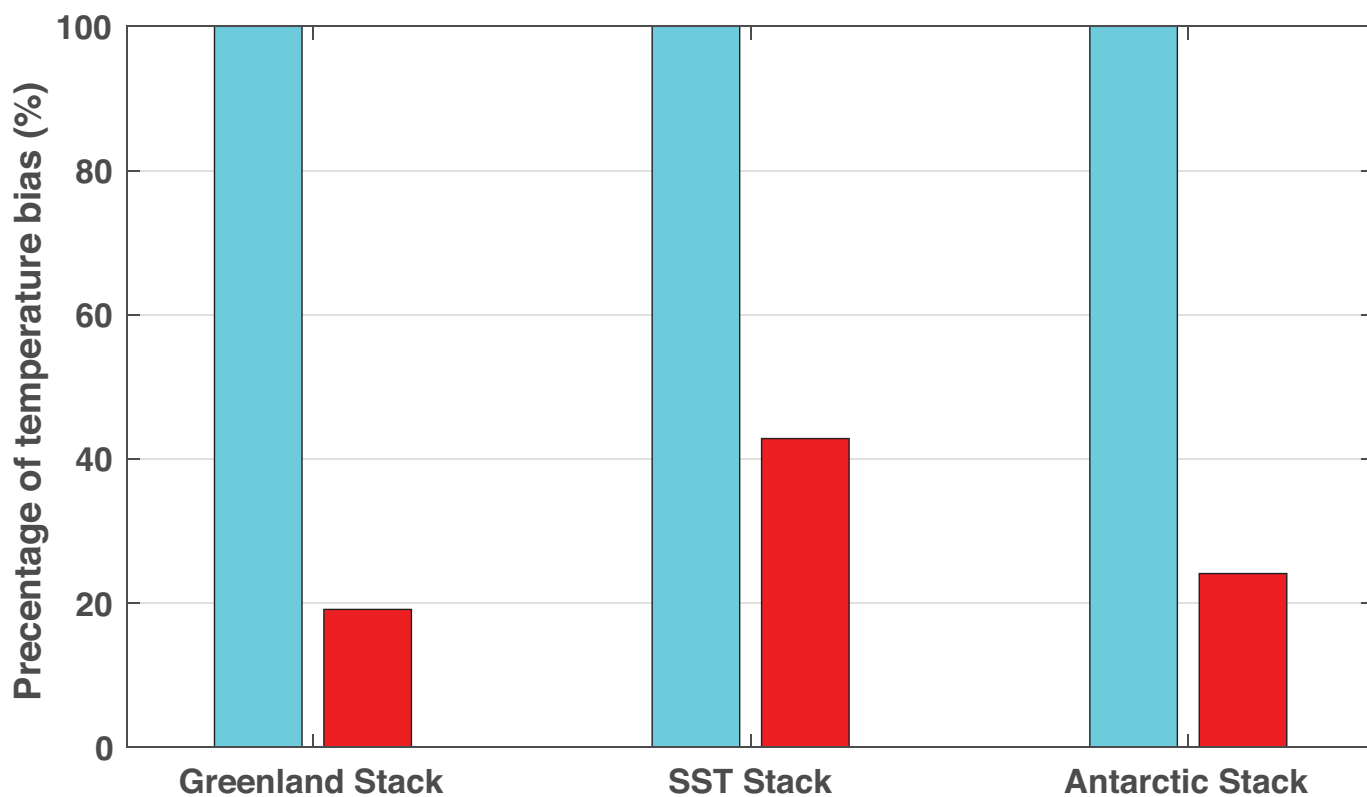
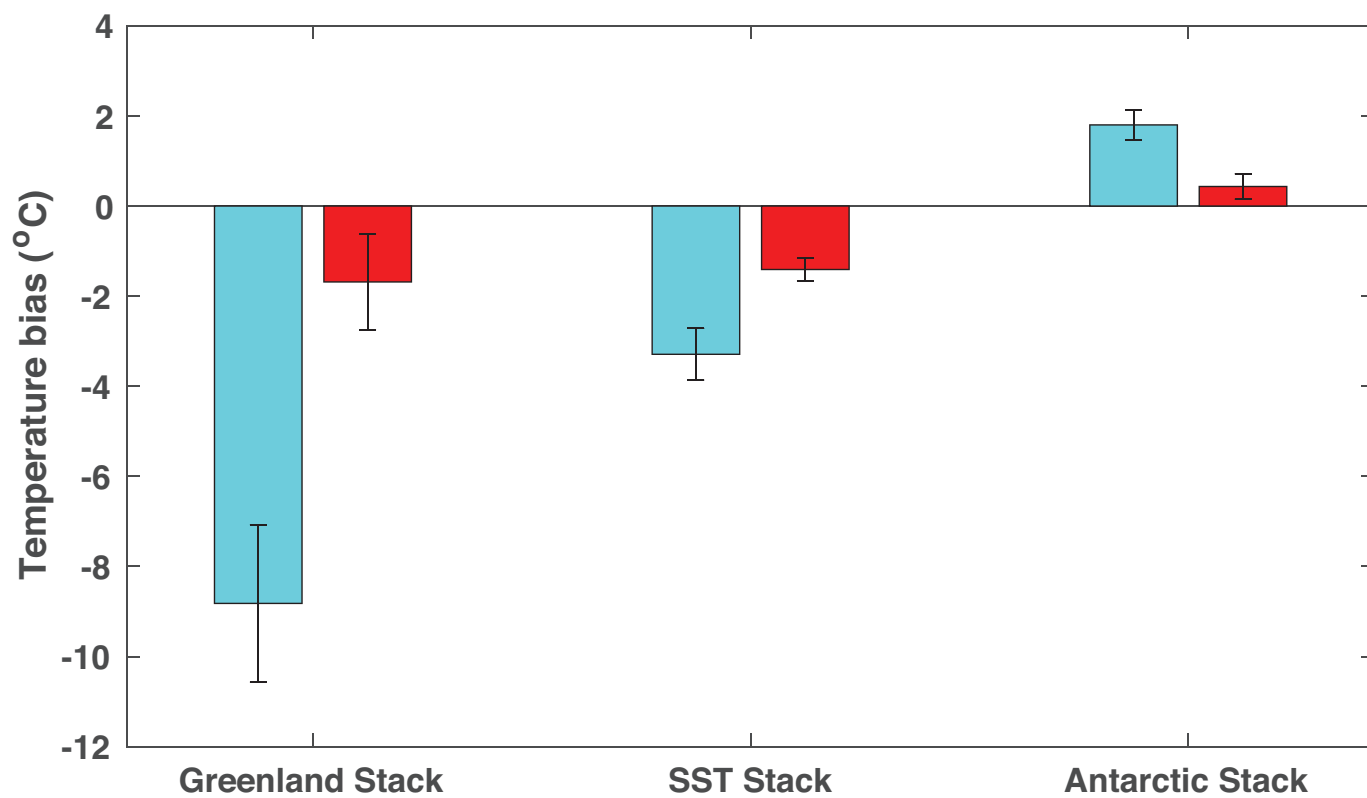


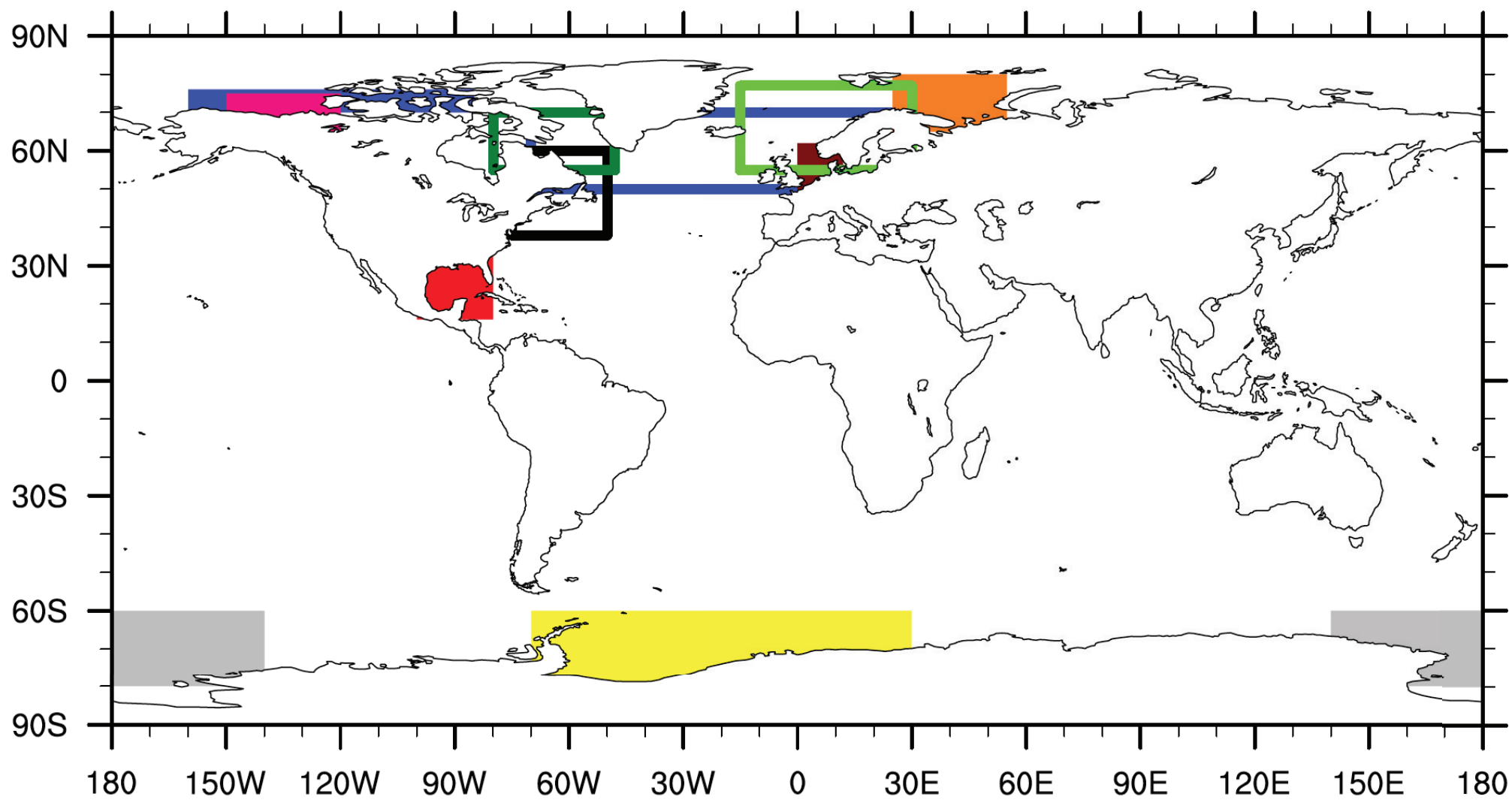


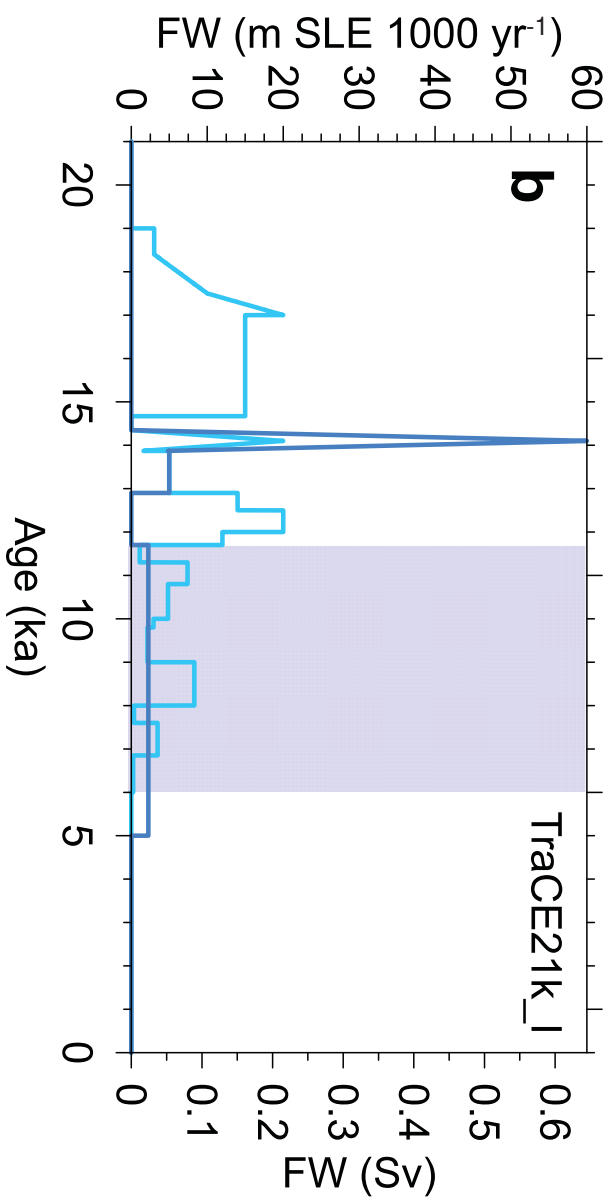
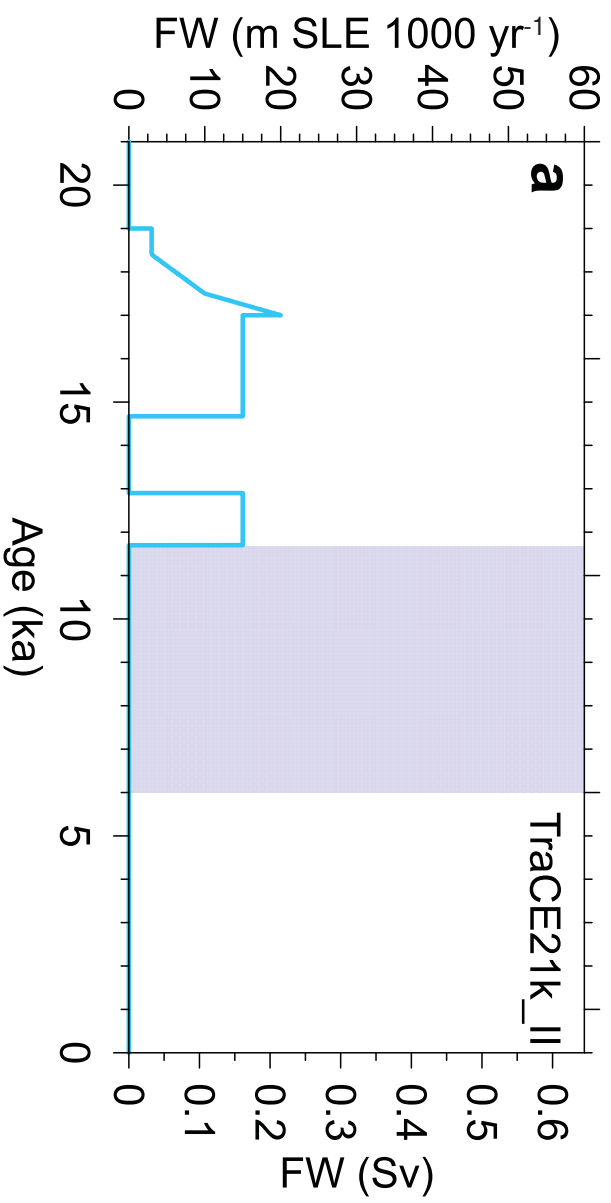




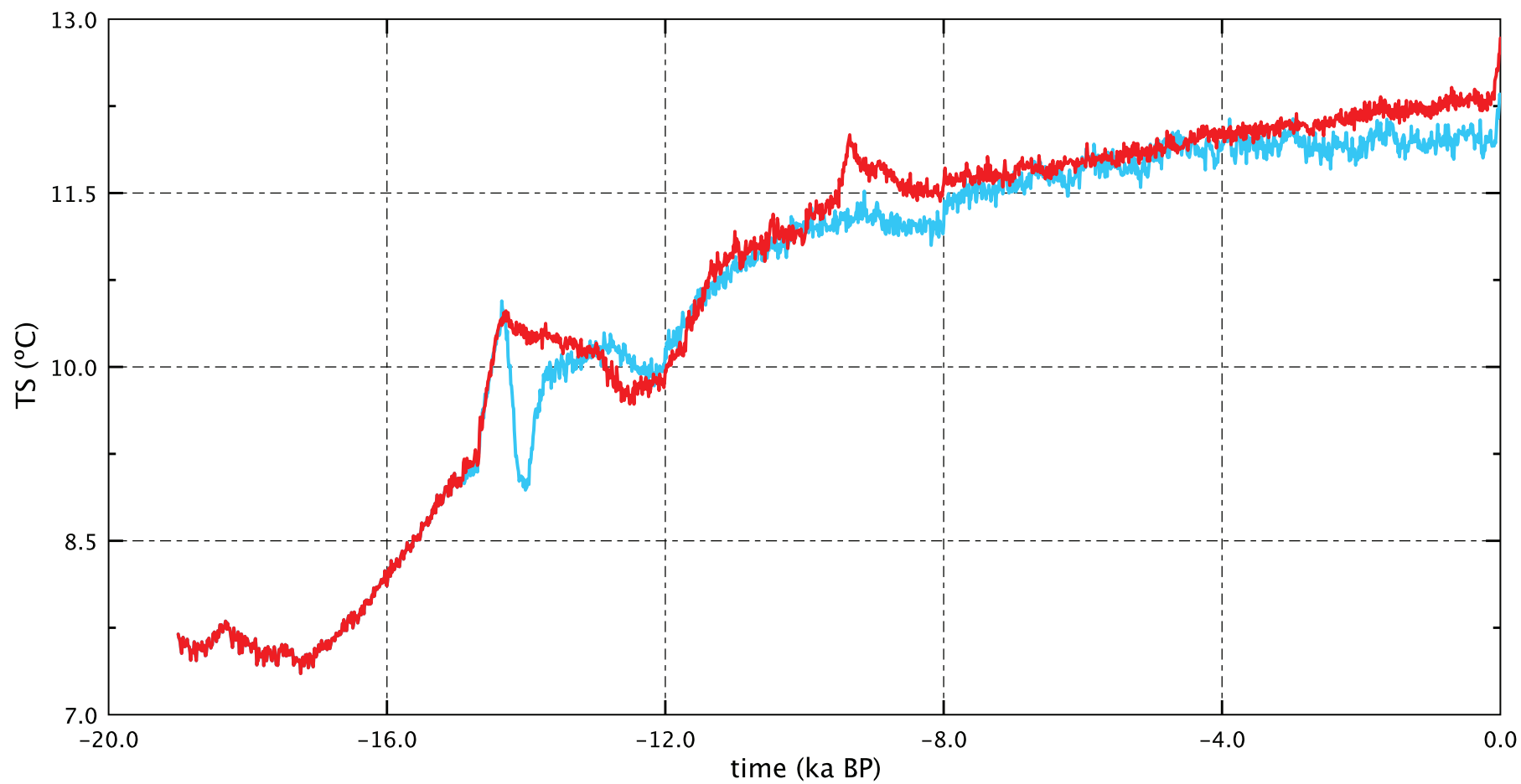








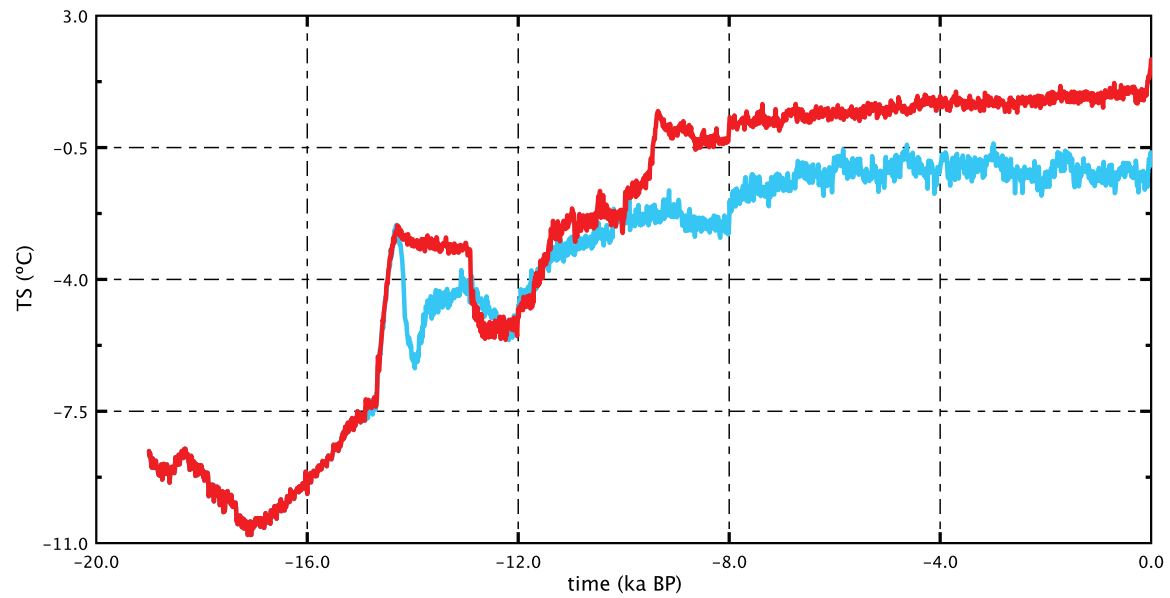
Global Mean



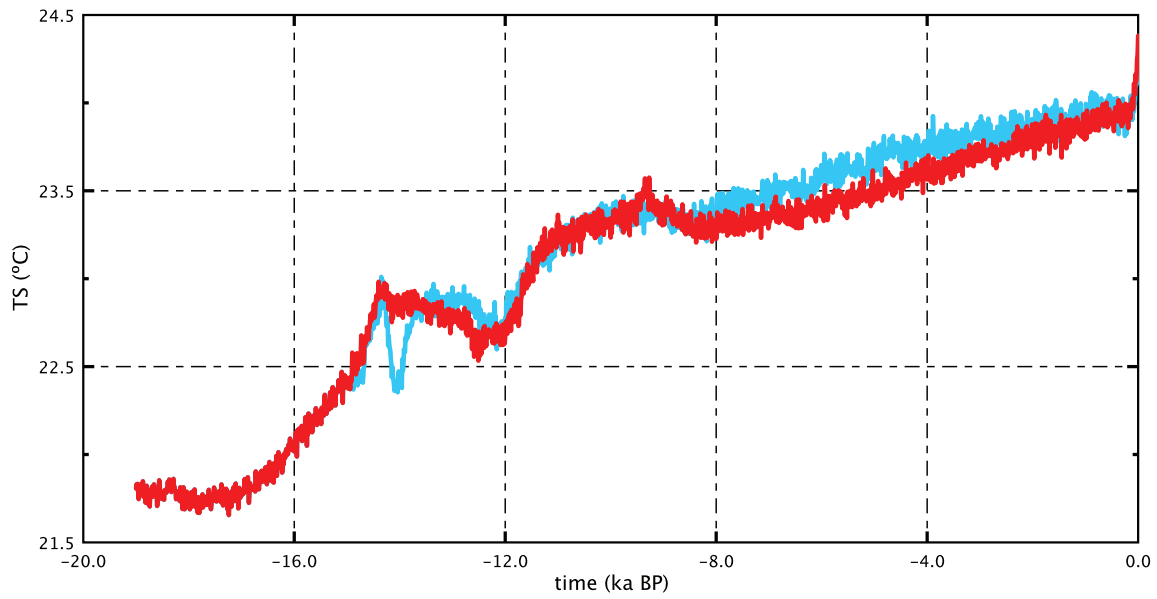
TraCE-21K-I

TraCE-21K-II

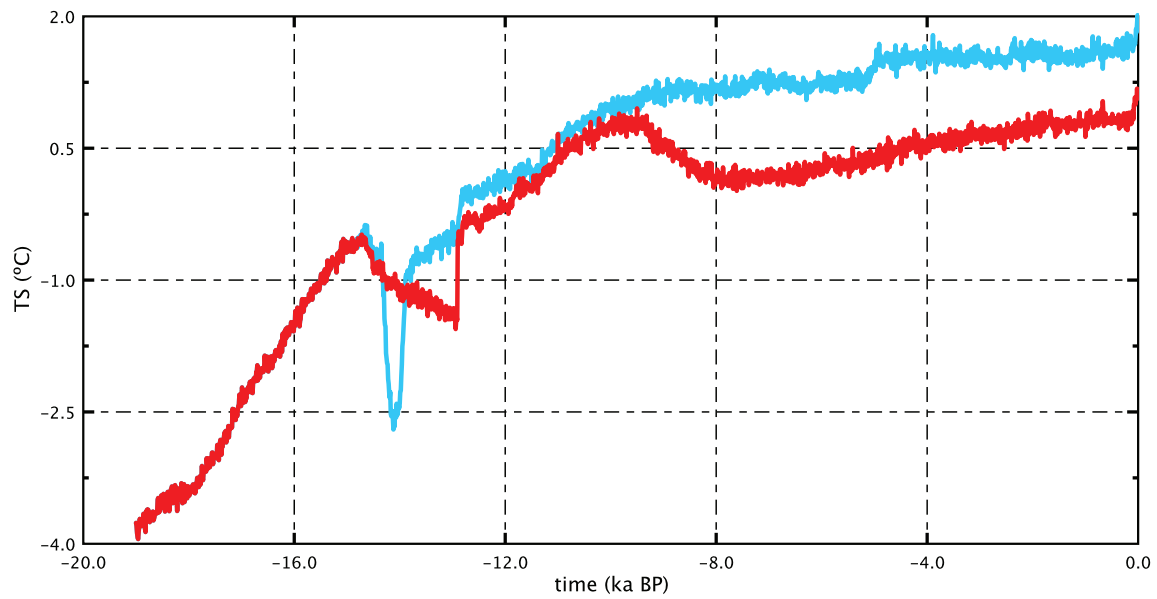
30N-90N Mean



30S-30N Mean



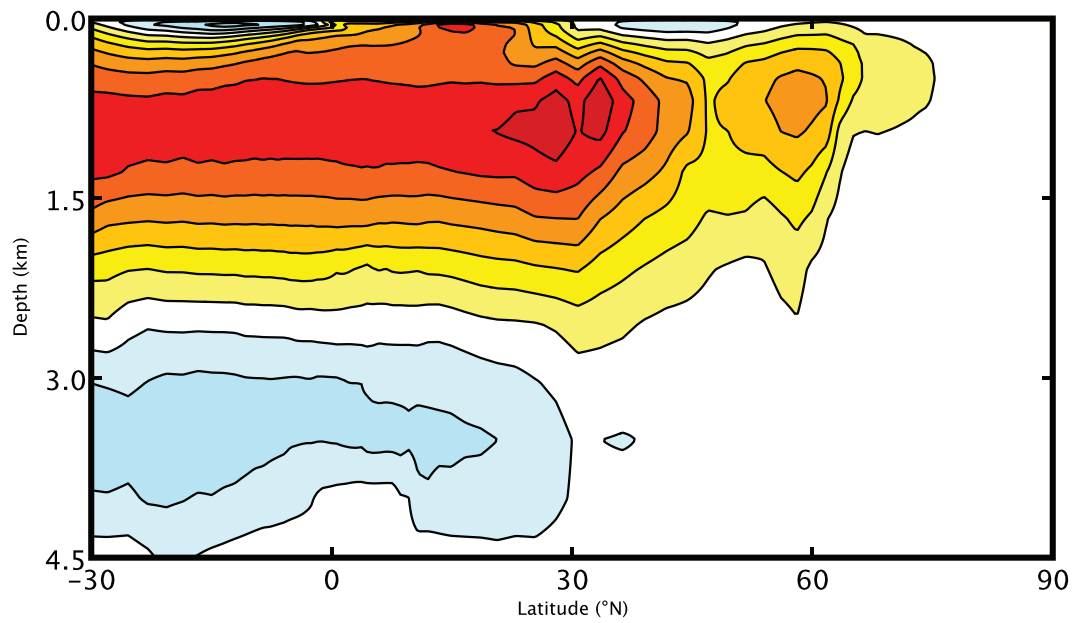
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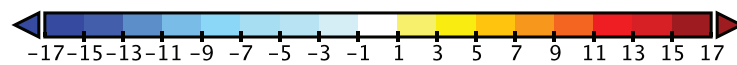
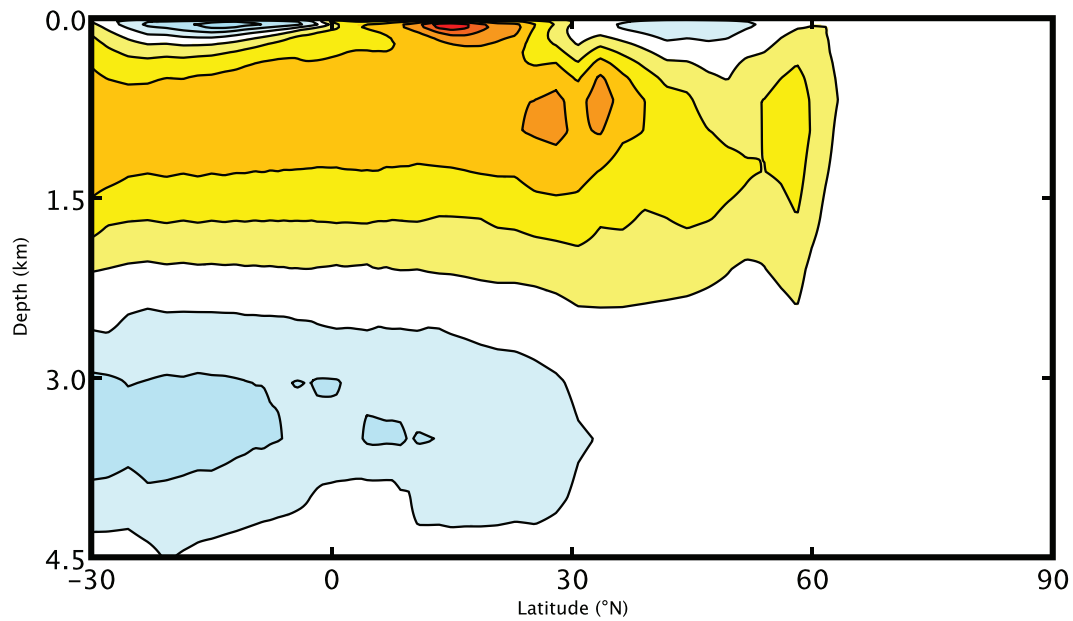
TraCE-21K-I

TraCE-21K-II

Meridional Overturning Circulation



Meridional Overturning Circulation



Maximum Mixed-Layer Depth

